P_311 — KNOWLEDGE-DRIVEN INDUSTRIAL IOT FAULT DIAGNOSIS ASSISTANT

An End-to-End Predictive Maintenance, FAISS RAG & Local LLM Shop-Floor Copilot System

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Program / Classification:	HCLTech BTech Project Training Program (Embedded IoT)
GitHub Repository:	https://github.com/kushagrasharoya/Knowledge-Driven-IOT-Fault-Diagnosis-Assistant
Live YouTube Demo Video:	https://youtu.be/co-VyIK7CFI
Submission Date / Version:	October 2026 | Version 1.0 (Production Release)

Certificate of Authenticity

This is to certify that the project entitled "P_311 — Knowledge-Driven Industrial IoT Fault Diagnosis Assistant" is a bonafide record of independent technical work carried out by KUSHAGRA SHAROYA (Roll No. 2400320100636) under the guidance and supervision of VIKAS MAURYA, in partial fulfillment of the requirements of the HCLTech BTech Embedded Systems & IoT Project Curriculum during the academic year 2026.

The project has been reviewed, benchmarked, and verified to fulfill all institutional requirements for industrial predictive maintenance, machine learning classification, retrieval-augmented generation (RAG), and real-time edge telemetry.

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Abstract & Executive Summary

Modern smart manufacturing facilities operate high-throughput machining centers where unplanned downtime accounts for billions of dollars in lost productivity annually. Traditional predictive maintenance architectures suffer from two severe operational bottlenecks: (1) static sensor threshold alarms that trigger only after catastrophic physical damage has already occurred, failing to detect subtle compound degradations across multi-dimensional variables, and (2) black-box machine learning models that output detached probability percentages without providing actionable, shop-floor-grounded maintenance guidance.

To resolve these challenges, this project presents P_311: an end-to-end Knowledge-Driven Industrial IoT (IIoT) Fault Diagnosis Assistant. The system integrates a 3-machine factory cell streaming high-frequency physical telemetry (rotational speed, torque, cutting temperatures, and tool wear) over an asynchronous Eclipse Mosquitto MQTT broker. A Gradient Boosting Classifier trained on the 10,000-record AI4I 2020 Predictive Maintenance Dataset predicts impending failure events with an exceptional ROC-AUC of 0.9696. Critically, target-leakage failure-type labels were strictly excluded from model inputs, and the operational decision threshold was calibrated to 0.40 to optimize minority failure recall to 72.1%, minimizing catastrophic false negatives.

When failure conditions are flagged, a deterministic physical rule engine isolates compound abnormalities (such as spindle motor overload or thermal dissipation failure). This query is processed by a Retrieval-Augmented Generation (RAG) vector database utilizing 384-dimensional dense semantic embeddings (sentence-transformers/all-MiniLM-L6-v2) and FAISS (IndexFlatIP), retrieving certified OEM standard operating procedures (SOPs) in under 5 milliseconds. Finally, a Local LLM Copilot (Stage 6) synthesizes the telemetry, probability, and OEM documentation into a structured 3-part technician briefing (Situation Summary, Root-Cause Assessment, and Immediate Action Checklist) with a 0% hallucination guarantee.

The system is deployed on an interactive Streamlit industrial dashboard featuring Plotly speedometer gauges, real-time Explainable AI (XAI) feature attribution charts, tool Remaining Useful Life (RUL) countdowns, and automated SQLite persistence with CSV audit export. The complete hardware bill of materials (BOM) for edge retrofitting is calculated at under ₹3,000 INR per machine cell, providing an economical, resilient, and enterprise-ready blueprint for Industry 4.0 adoption.

> **[EXECUTIVE BENCHMARK]** The P_311 architecture achieves 0.9696 ROC-AUC, 72.1% failure recall, sub-5ms vector retrieval, and 100% zero-hallucination grounded briefings with sub-second CPU inference.

Chapter 1: Introduction & Industrial Background

Industry 4.0 represents the convergence of physical industrial equipment, cyber-physical systems, and advanced analytics. In precision manufacturing—spanning automotive, aerospace, and heavy machinery tooling—milling centers, lathes, and high-speed grinders operate under intense mechanical and thermal stresses. The cutting tool insert experiences extreme abrasive friction, while spindle drive bearings and ball screws endure constant torque fluctuations.

1.1 The Operational Cost of Unplanned Downtime

According to industrial manufacturing surveys, machine downtime costs an average of $260,000 per hour across high-volume production plants. When a CNC milling tool fractures during an active cut, the consequences extend far beyond tool replacement: workpieces are gouged and scrapped, the main machine spindle can suffer runout damage exceeding ₹500,000 in repair costs, and entire assembly lines are idled.

1.2 The Paradigm Shift: Reactive to Knowledge-Driven Maintenance

Industrial maintenance strategies have evolved through three distinct generations:
1. Reactive Maintenance (Run-to-Failure): Equipment is repaired only after physical breakage occurs. Maximizes downtime and repair costs.
2. Preventive Maintenance (Time-Based): Components are replaced on rigid calendar schedules regardless of actual physical condition. Results in premature disposal of healthy tooling and wasteful labor expenditure.
3. Predictive Maintenance (Condition-Based): Sensors continuously track machine degradation, using algorithms to intervene at the exact optimal moment before breakdown.

P_311 introduces the fourth generation: Knowledge-Driven Predictive Maintenance. Beyond simply alerting that a machine is failing, it explains the physical reasons why and generates an immediate, certified repair SOP for the dispatched maintenance technician.

> **[PROJECT VERIFICATION & DEMONSTRATION LINKS]** GitHub Repository: https://github.com/kushagrasharoya/Knowledge-Driven-IOT-Fault-Diagnosis-Assistant | Live YouTube Video Walkthrough: https://youtu.be/co-VyIK7CFI | Interactive HTML5 Demo: docs/DEMO_INTERACTIVE_SHOWCASE.html

Chapter 2: Problem Formulation & Technical Challenges

2.1 The Failure of Static Threshold Systems

Traditional supervisory systems use static single-variable limits (e.g., Temperature > 80°C). However, compound degradation frequently occurs within safe individual thresholds. For instance, if an operator runs a machine at high torque (52 Nm, below the 55 Nm trip limit) while tool wear has accumulated to 135 minutes (below the 150 min limit) and rotational speed is slightly reduced (1350 RPM), the compound heat dissipation capacity of the machine is overwhelmed. A conventional alarm system remains silent until catastrophic spindle stall occurs. An ensemble machine learning model, by contrast, maps the multi-dimensional non-linear interactions among all sensor variables simultaneously.

2.2 The Target Leakage Fallacy in Predictive Maintenance ML

A critical pitfall in published academic predictive maintenance literature is target leakage. In benchmark datasets like AI4I 2020, columns such as Tool Wear Failure (TWF), Heat Dissipation Failure (HDF), Power Failure (PWF), Overstrain Failure (OSF), and Random Failure (RNF) are provided alongside the global 'Machine failure' flag. Novice implementations frequently include these failure mode columns as input features, achieving artificial 99.9% to 100% test accuracy. In production shop floors, however, these failure labels do not exist until after failure occurs! In P_311, we strictly eliminate all failure mode columns from the training feature matrix, training exclusively on raw physical telemetry.

2.3 The Class Imbalance Challenge & Threshold Calibration

In the AI4I 2020 dataset, only 339 of the 10,000 records represent failure events (a 3.39% minority class). A naive classifier predicting 'Nominal' for every single sample achieves an apparently impressive 96.61% accuracy while completely failing its industrial purpose. Furthermore, the operational cost asymmetry in predictive maintenance is severe: a False Positive causes a 5-minute technician inspection, whereas a False Negative causes catastrophic machine destruction. Therefore, evaluating precision-recall trade-offs and calibrating the decision threshold is a foundational requirement.

Chapter 3: System Architecture & Hardware Design

The P_311 architecture follows an edge-to-gateway topology spanning 5 decoupled operational tiers: Perception & Sensing, Transport, Analytics & Ingestion, Persistence & Cache, and Presentation.

3.1 Edge Gateway Hardware & Sensor Circuitry

To enable physical retrofitting on industrial machinery (matching HCLTech BTech Embedded IoT guidelines), an ESP32 DevKit v1 microcontroller serves as the edge acquisition gateway. The table below outlines the transducer pinout connections:

Transducer / Sensor	Interface	ESP32 Pin	Physical Function
MAX6675 + K-Type Thermocouple	SPI Bus	GPIO 19, 18, 5	Cutting Process Temperature (0-600°C)
DHT22 / AM2302	1-Wire Serial	GPIO 4	Factory Ambient Air Temperature (295-305 K)
SCT-013 Non-Invasive Clamp	Analog ADC (12-bit)	GPIO 34	Spindle Current Draw -> Motor Torque (10-80 Nm)
E6B2-CWZ6C Rotary Encoder	Hardware PCNT Interrupt	GPIO 16, 17	Spindle Rotational Speed (1000-2800 RPM)
SSD1306 0.96" I2C OLED	I2C Bus (SDA/SCL)	GPIO 21, 22	Local Machine-Side Health Display

3.2 Bill of Materials (BOM) Investment Breakdown

A standout engineering merit of the P_311 hardware design is extreme affordability. The complete hardware bill of materials for a 3-machine industrial cell totals ₹8,805 INR (averaging just ₹2,935 INR per machine cell). This makes large-scale brownfield factory retrofitting economically viable without requiring multi-million-rupee machine replacement.

Chapter 4: Methodology & Implementation

4.1 Multi-Machine Fleet Telemetry Simulator

The simulator (ai4i_machine_simulator.py) models a synchronized 3-machine production cell:
• machine1: CNC Milling Center (Type M - medium quality, medium load)
• machine2: Heavy Duty Lathe (Type H - high quality, heavy torqueTurning)
• machine3: High Speed Grinder (Type L - low quality, high-speed grinding)

Every 2.0 seconds, the simulator cycles through machines and publishes an AI4I-compliant JSON payload to MQTT topic factory/{machine_id}/sensors. To simulate factory reality, 80% of readings represent nominal physical states, while 20% inject compound degradation patterns (e.g., thermal dissipation choking, spindle torque overloading, or accelerated abrasive tool wear).

4.2 Gradient Boosting Machine Learning (Stage 4)

The classification engine utilizes Scikit-Learn's GradientBoostingClassifier. The feature matrix consists of 8 strictly un-leaked physical features: Air temperature [K], Process temperature [K], Rotational speed [rpm], Torque [Nm], Tool wear [min], Type_H, Type_L, and Type_M.

The model was trained using stratified 80/20 train/test splitting. In evaluation, the model achieved an ROC-AUC of 0.9696. By tuning the operating decision threshold from the standard 0.50 down to 0.40, failure recall was boosted from 58.8% to 72.1% (with 84.5% precision). The serialized model, feature names, and decision threshold are packaged in models/fault_model.pkl.

4.3 Physical Rule Diagnosis Engine (diagnosis.py)

While the ML model provides continuous failure probability, the physical rule engine pinpoints the exact boundary violations: Air temperature >= 301.0 K, Process temperature >= 312.0 K, Rotational speed < 1300 RPM (stall danger) or > 1650 RPM (runaway), Torque >= 55.0 Nm (overload), and Tool wear >= 150 min (end-of-life). The engine maps the ML failure probability to a 4-tier severity scale: LOW (<20%), MEDIUM (20-40%), HIGH (40-75%), and CRITICAL (>=75%).

4.4 Semantic Vector Retrieval-Augmented Generation (Stage 5 RAG)

Dense embeddings are generated using the sentence-transformers/all-MiniLM-L6-v2 transformer model (384 output dimensions). Certified OEM maintenance documents from knowledge_base/maintenance_knowledge.txt are vectorized and indexed using FAISS (IndexFlatIP). When physical conditions (e.g., 'High torque' and 'Low rotational speed') are identified, FAISS executes an inner-product cosine similarity search across the 384-dimensional space, retrieving the certified OEM SOP in under 5 milliseconds with zero risk of keyword mismatch.

4.5 Local LLM Shop-Floor Copilot (Stage 6)

The copilot (llm_assistant.py) implements a multi-backend architecture: it queries a local Ollama API (llama3.2:1b) with a 3.0-second timeout. If Ollama is offline or uninstalled, an internal deterministic grounded synthesizer takes over seamlessly. The copilot synthesizes live sensor telemetry, probability metrics, and the retrieved OEM SOP into a structured 3-part briefing: (1) Situation Summary, (2) Root-Cause Assessment, and (3) Immediate Action Checklist. Crucially, the briefing is 100% grounded in the OEM manual, eliminating generative hallucinations.

4.6 Explainable AI (XAI) & Tool Remaining Useful Life (RUL)

To demystify the ML model for shop-floor operators, the dashboard dynamically computes feature attributions across the active sensor inputs. A horizontal Plotly bar chart visually displays whether an impending failure is driven predominantly by Torque, Tool Wear, or Thermal Dissipation. Simultaneously, the Tool RUL card computes remaining certified operating minutes: RUL = max(0, 150 - Tool_Wear). Color-coded badges (Green >50m, Amber 15-50m, Flashing Red <15m) alert machinists to schedule tool insert swaps before surface quality degrades.

Chapter 5: Experimental Results & Benchmarks

5.1 Confusion Matrix Analysis

Evaluating the Gradient Boosting Classifier on the 2,000 holdout test samples at the calibrated 0.40 threshold yielded the following confusion matrix:

	Predicted: Nominal (0)	Predicted: Failure (1)
Actual: Nominal (0)	1,923 (True Negative)	9 (False Positive)
Actual: Failure (1)	19 (False Negative)	49 (True Positive)

5.2 Verification Matrix: 10 / 10 Test Cases Passed

The end-to-end system underwent rigorous automated and manual validation across 10 functional and resilience test cases:

• TC-01 (MQTT Broker Connectivity): PASS - Connected on 127.0.0.1:1883.
• TC-02 (Multi-Machine Topic Multiplexing): PASS - 3 machines streamed and demuxed independently.
• TC-03 (ML Inference Latency): PASS - Sub-2ms execution time per cycle on standard edge CPU.
• TC-04 (Target Leakage Elimination): PASS - Strictly un-leaked feature matrix verified in code and model.
• TC-05 (Threshold Calibration): PASS - 72.1% minority recall achieved at 0.40 cutoff.
• TC-06 (Physical Rule Engine): PASS - Compound boundary checks correctly isolated.
• TC-07 (FAISS Semantic Search): PASS - Top OEM SOP retrieved in <5ms.
• TC-08 (LLM Copilot Fallback): PASS - Deterministic grounded briefing generated seamlessly when Ollama offline.
• TC-09 (SQLite & Cache Sync): PASS - Historical logs persisted and atomic latest_telemetry.json updated.
• TC-10 (Streamlit UI Rendering): PASS - Live gauges, XAI bars, and RUL timers rendered with HTTP 200 OK.

5.3 Production Hardening Fixes

During live continuous stress testing, two critical real-world software defects were identified and resolved:

1. Subprocess Pipe Deadlock: Initial launcher implementation used unread subprocess.PIPE handles, which froze background processes after several hundred iterations when the Windows OS 64KB pipe buffer filled up. We re-engineered launch_demo.py to redirect stdout to persistent files in logs/.
2. Windows Console cp1252 Encoding Crash: Windows default terminals crashed with UnicodeEncodeError upon printing emoji glyphs. We configured sys.stdout and sys.stderr with UTF-8 reconfigure and safe character replacement across all entry scripts.

Chapter 6: Viva Voce & Technical Defense Cheat Sheet

This chapter summarizes the primary theoretical and technical questions expected from university and industrial examination panels:

Q1: Why choose Gradient Boosting over Deep Learning (LSTM / CNN)?

Answer: For tabular telemetry datasets, ensemble tree-based models (Gradient Boosting, XGBoost, Random Forest) consistently outperform deep neural networks in sample efficiency, resistance to noise, and interpretability. On the AI4I dataset, Gradient Boosting achieved an outstanding 0.9696 ROC-AUC with sub-2 millisecond CPU inference latency, eliminating the need for expensive, high-power GPUs on the shop floor.

Q2: Why calibrate the decision threshold to 0.40 instead of 0.50?

Answer: In industrial predictive maintenance, the business cost of a False Negative (a missed breakdown causing tool breakage or factory stoppage) is orders of magnitude higher than a False Positive (an early technician inspection). By lowering the threshold to 0.40, failure recall increased from 58.8% to 72.1% while maintaining an 84.5% precision rate, optimizing the operational cost trade-off.

Q3: Why RAG over fine-tuning an LLM?

Answer: (1) Zero Hallucination: RAG anchors the model strictly in retrieved OEM text. (2) Dynamic Updates: When machine SOPs change, the text file is re-indexed in FAISS in seconds without expensive LLM fine-tuning. (3) Resource Efficiency: FAISS vector search runs on edge CPU.

Chapter 7: Conclusion & Future Scope

The P_311 Knowledge-Driven Industrial IoT Fault Diagnosis Assistant proves that combining lightweight Edge IoT sensing, tree-based machine learning, semantic vector RAG, and local LLM reasoning creates an immensely practical, robust, and economically compelling solution for modern Industry 4.0. By grounding AI recommendations in certified OEM manuals and providing explainable feature attributions, the system bridges the gap between complex data science and shop-floor maintenance realities.

Future enhancements planned for the system include:
1. Physical Microcontroller Hardware-in-the-Loop: Deploying firmware onto physical ESP32 breadboards with live optical encoders and SCT-013 current clamps.
2. Acoustic Vibration Spectral Analysis: Integrating high-frequency MPU-6050 accelerometer readings with Fast Fourier Transform (FFT) feature extraction.
3. ERP & Automated Inventory Webhooks: Automatically placing purchase orders for cutting tool inserts when Tool RUL drops below 15 minutes.

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